INCH-POUND
MIL-P-23469/7B(SH)
19 February 1988
SUPERSEDING
MIL-P-23469/7A(SHIPS)
31 May 1967

MILITARY SPECIFICATION SHEET

Ⓑ

PIN-RIVET, GROOVED, OVAL 60° COUNTERSUNK HEAD; TAPERED-SECTION SHANK, MULTIPLE LOCKING GROOVES, ALUMINUM ALLOY, CORROSION-RESISTANT STEEL AND CARBON STEELS

This specification sheet is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

Ⓑ The requirements for acquiring the pins described herein shall consist of this specification and the latest issue of MIL-P-23469.

Oval 60 degree countersunk head, swage-locking pins shall be furnished in the following type and classes. Pins manufactured to this specification are intended for use with collars manufactured to MIL-P-23469.

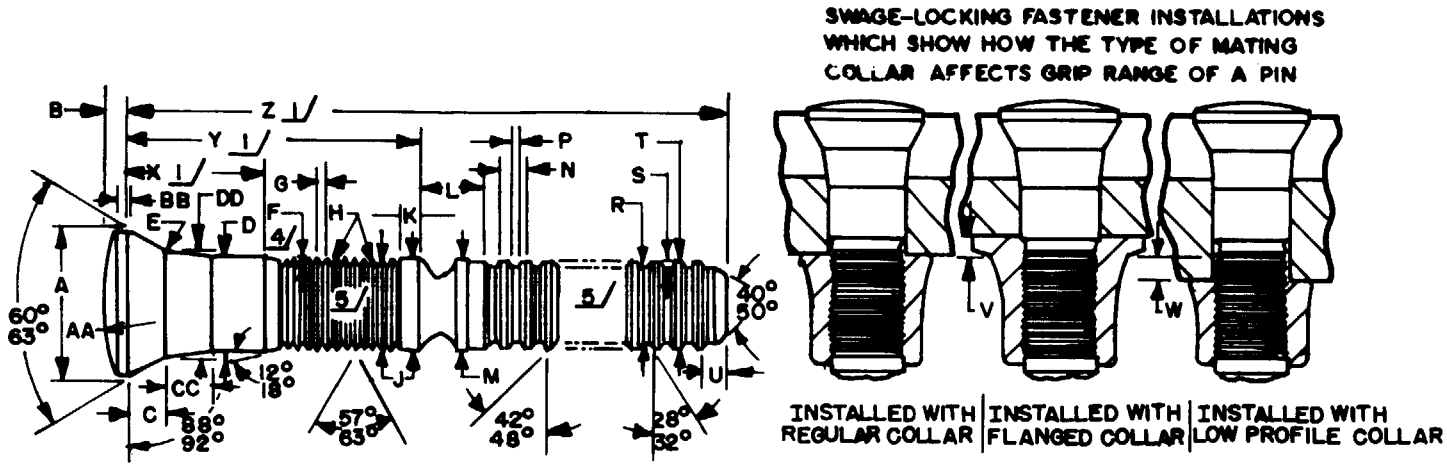
Type II - multiple locking grooves (see figure 1)

Class 1 - Aluminum alloy
Class 2 - Corrosion resistant steel
Class 3 - Carbon steel

Ⓑ Denotes changes.

AMSC N/A FSC 5320
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SH 13203044

Nom dia	A Dia 2/	AA 2/	AAA 3/	B 2/	BB Max	C 2/	CC	D Dia	DD Dia	E Rad	F	G Pitch	H Rad	J Dia	K Max	L Max
1/2	.808 .768	.990 1.010	.030	.078 .093	.031	.218 .238	.219 .281	.510 .493	.530 .520	.020 .040	.025 .028	.039 .045	.002 .010	.495 .485	.094	.375
5/8	1.005 .965	1.115 1.135	.040	.102 .117	.047	.274 .294	.281 .344	.635 .617	.658 .648	.025 .050	.031 .035	.052 .058	.003 .013	.618 .608	.115	.375
3/4	1.203 1.163	1.365 1.385	.047	.125 .140	.062	.337 .357	.344 .406	.760 .741	.780 .770	.030 .060	.036 .040	.059 .065	.003 .015	.740 .730	.135	.500
10/ 7/8	1.400 1.360	1.615 1.635	.055	.149 .164	.078	.393 .413	.406 .469	.885 .866	.905 .895	.035 .065	.043 .048	.068 .076	.003 .017	.870 .860	.156	.625
10/ 1	1.597 1.557	1.803 1.823	.062	.172 .187	.094	.454 .474	.469 .531	1.010 .990	1.030 1.020	.040 .070	.052 .058	.080 .088	.004 .020	.993 .983	.180	.625

See footnotes at end of table.

FIGURE 1. Type II - multiple locking grooves.

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Nom dia	M Dia	N Pitch	P	R Dia	S	T Max dia	U Min	V 7/ Flanged collar	W 8/ Low profile collar	Hole preparation data		
										Size drill	Max hole dia	C'sink dia
1/2	.473 .449	.0615 .0635	.016 .018	.427 .423	.020 .023	.473	.062	.125	.140	1/2 (.500)	33/64	23/32
5/8	.615 .570	.0771 .0791	.020 .023	.530 .524	.026 .030	.590	.078	.156	.180	5/8 (.625)	41/64	15/16
3/4	.740 .682	.0927 .0947	.025 .028	.637 .631	.032 .037	.711	.094	.188	.215	3/4 (.750)	49/64	1-1/8
10/ 7/8	.860 .790	.1084 .1104	.029 .032	.746 .738	.036 .041	.828	.109	.218	.250	7/8 (.875)	57/64	1-1/4
10/ 1	.980 .900	.124 .126	.032 .036	.857 .849	.041 .047	.951	.125	.250	.285	1 (1.000)	1-1/64	1-7/16

1/ "X", "Y", and "Z" shown in table I.
2/ Dimensions "A", "AA", "B", "C", "X", "Y" and "Z" are from theoretical intersection.
3/ Conical surface of head to "D" diameter shall be concentric within "AAA" TIR.
4/ Initial locking groove formed at juncture of angle and "D" body diameter.
5/ Dimensions indicated on locking grooves and pull grooves are typical.
6/ Accumulated pitch tolerance not to exceed plus or minus 0.002 on any 10 consecutive pull grooves.

Collar: Regular height - use grip shown in table I.

7/ Flanged - add flange thickness "V" to actual thickness to be fastened to determine "grip" to be used in specifying dash number, table I.
8/ Low profile - subtract "W" from actual thickness to be fastened to determine "grip" to be used in specifying dash number, table I.
9/ Regular height collar required for underwater application.
10/ 7/8 inch and 1 inch not available in ALUM and CRES.

NOTES:
1. MIL-P-23469/7B and MIL-P-23469/7A pins are interchangeable type for type, class for class, and size for size.
2. Dimensions are in inches.
3. Shank shall be straight within 0.004 TIR per inch of length.
4. 60/63 degree angle and "D" diameter to be concentric within 0.005 TIR.

FIGURE 1. Type II - multiple locking grooves. - Continued

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TABLE I. Multiple locking grooves.

Dash number identification																
M23469/7 1/			1/2 diameter					M23469/7 1/			3/4 diameter				
ALUM Class 1	CRES Class 2	Carbon steel Class 3	Grip range Min	Grip range Max	X +.016 -.031	Y ±.020	Z +.00 -.26	ALUM Class 1	CRES Class 2	Carbon steel Class 3	Grip range Min	Grip range Max	X +.016 -.031	Y ±.020	Z +.00 -.26
-11612	-21612	-31612	.751	1.000	.688	1.713	3.86								
-11616	-21616	-31616	1.001	1.250	.938	1.963	4.11	-12416	-22416	-32416	1.001	1.250	.938	2.268	5.09
-11620	-21620	-31620	1.251	1.500	1.188	2.213	4.36	-12420	-22420	-32420	1.251	1.500	1.188	2.518	5.34
-11624	-21624	-31624	1.501	1.750	1.438	2.463	4.61	-12424	-22424	-32424	1.501	1.750	1.438	2.768	5.59
-11628	-21628	-31628	1.751	2.000	1.688	2.713	4.86	-12428	-22428	-32428	1.751	2.000	1.688	3.018	5.84
-11632	-21632	-31632	2.001	2.250	1.938	2.963	5.11	-12432	-22432	-32432	2.001	2.250	1.938	3.268	6.09
-11636	-21636	-31636	2.251	2.500	2.188	3.213	5.36	-12436	-22436	-32436	2.251	2.500	2.188	3.518	6.34
-11640	-21640	-31640	2.501	2.750	2.438	3.463	5.61	-12440	-22440	-32440	2.501	2.750	2.438	3.768	6.59
-11644	-21644	-31644	2.751	3.000	2.688	3.713	5.86	-12444	-22444	-32444	2.751	3.000	2.688	4.018	6.84
-11648	-21648	-31648	3.001	3.250	2.938	3.963	6.11	-12448	-22448	-32448	3.001	3.250	2.938	4.268	7.09
-11652	-21652	-31652	3.251	3.500	3.188	4.213	6.36	-12452	-22452	-32452	3.251	3.500	3.188	4.518	7.34
-11656	-21656	-31656	3.501	3.750	3.438	4.463	6.61	-12456	-22456	-32456	3.501	3.750	3.438	4.768	7.59
-11660	-21660	-31660	3.751	4.000	3.688	4.713	6.86	-12460	-22460	-32460	3.751	4.000	3.688	5.018	7.84
-11664	-21664	-31664	4.001	4.250	3.938	4.963	7.11	-12464	-22464	-32464	4.001	4.250	3.938	5.268	8.09

See footnote at end of table.

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TABLE I. Multiple locking grooves. - Continued

Dash number identification																
M23469/7 1/				5/8 diameter					M23469/7 1/			7/8 diameter				
ALUM	CRES	Carbon steel	Carbon steel high strength	Grip range		X	Y	Z			Carbon steel high strength	Grip range		X	Y	Z
Class 1	Class 2		Class 5	Min	Max	+.016 -.031	±.020	+.00 -.26			Class 5	Min	Max	+.016 -.031	±.020	+.00 -.26
-12012	-22012		-52012	.751	1.000	.688	1.863	4.28								
-12016	-22016		-52016	1.001	1.250	.938	2.113	4.53			-52816	1.001	1.250	.938	2.421	5.38
-12020	-22020		-52020	1.251	1.500	1.188	2.363	4.78			-52820	1.251	1.500	1.188	2.671	5.63
-12024	-22024		-52024	1.501	1.750	1.438	2.613	5.03			-52824	1.501	1.750	1.438	2.921	5.88
-12028	-22028		-52028	1.751	2.000	1.688	2.863	5.28			-52828	1.751	2.000	1.688	3.171	6.13
-12032	-22032		-52032	2.001	2.250	1.938	3.113	5.53			-52832	2.001	2.250	1.938	3.421	6.38
-12036	-22036		-52036	2.251	2.500	2.188	3.363	5.78			-52836	2.251	2.500	2.188	3.671	6.63
-12040	-22040		-52040	2.501	2.750	2.438	3.613	6.03			-52840	2.501	2.750	2.438	3.921	6.88
-12044	-22044		-52044	2.751	3.000	2.688	3.863	6.28			-52844	2.751	3.000	2.688	4.171	7.13
-12048	-22048		-52048	3.001	3.250	2.938	4.113	6.53			-52848	3.001	3.250	2.938	4.421	7.38
-12052	-22052		-52052	3.251	3.500	3.188	4.363	6.78			-52852	3.251	3.500	3.188	4.671	7.63
-12056	-22056		-52056	3.501	3.750	3.438	4.613	7.03			-52856	3.501	3.750	3.438	4.921	7.88
-12060	-22060		-52060	3.751	4.000	3.688	4.863	7.28			-52860	3.751	4.000	3.688	5.171	8.13
-12064	-22064		-52064	4.001	4.250	3.938	5.113	7.53			-52864	4.001	4.250	3.938	5.421	8.38

See footnote at end of table.

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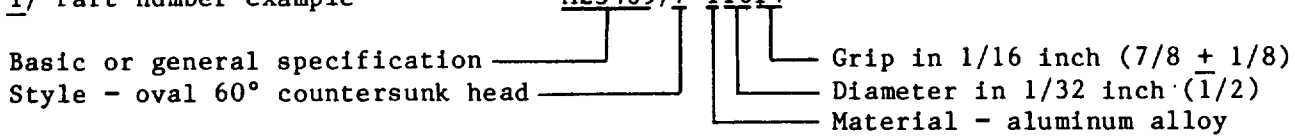
TABLE I. Multiple locking grooves. - Continued

Dash number identification																
M23469/7 1/				5/8 diameter					M23469/7 1/			1" diameter				
		Carbon steel		Grip range		X	Y	Z			Carbon steel	Grip range		X	Y	Z
		Class 3		Min	Max	+.016 -.031	±.020	+.00 -.26			Class 3	Min	Max	+.016 -.031	±.020	+.00 -.26
		-32012		.751	1.000	.688	1.933	4.34								
		-32016		1.001	1.250	.938	2.183	4.59								
		-32020		1.251	1.500	1.188	2.433	4.84			-33220	1.251	1.500	1.188	2.823	6.32
		-32024		1.501	1.750	1.438	2.683	5.09			-33224	1.501	1.750	1.438	3.073	6.57
		-32028		1.751	2.000	1.688	2.933	5.34			-33228	1.751	2.000	1.688	3.323	6.82
		-32032		2.001	2.250	1.938	3.183	5.59			-33232	2.001	2.250	1.938	3.573	7.07
		-32036		2.251	2.500	2.188	3.433	5.84			-33236	2.251	2.500	2.188	3.823	7.32
		-32040		2.501	2.750	2.438	3.683	6.09			-33240	2.501	2.750	2.438	4.073	7.57
		-32044		2.751	3.000	2.688	3.933	6.34			-33244	2.751	3.000	2.688	4.323	7.82
		-32048		3.001	3.250	2.938	4.183	6.59			-33248	3.001	3.250	2.938	4.573	8.07
		-32052		3.251	3.500	3.188	4.433	6.84			-33252	3.251	3.500	3.188	4.823	8.32
		-32056		3.501	3.750	3.438	4.683	7.09			-33256	3.501	3.750	3.438	5.073	8.57
		-32060		3.751	4.000	3.688	4.933	7.34			-33260	3.751	4.000	3.688	5.323	8.82
		-32064		4.001	4.250	3.938	5.183	7.59			-33264	4.001	4.250	3.938	5.573	9.07

See footnote at top of next page.

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1/ Part number example - M23469/7-11614



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